

Quantum (OneBitHHL) vs. classical sparse solver at the segment level

in a planar LHCb VELO toy: a direct head-to-head

Internal report — Quantum Track Reconstruction working note
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Abstract

We compare the classical sparse solve and the ONEBITHHL quantum filter on the same segment-level Hopfield Hamiltonian over the multiplicity sweep $T \in [2, 1000]$ of the §14e operating point of the planar LHCb VELO toy. Solver inputs — segment graph, $\gamma, \delta, \varepsilon, \tau$, detector resolutions, hit-drop fraction, and reconstructable-track validation cuts — are identical on both sides. We report segment efficiency, false rate, solution fidelity, and downstream tracking efficiency and ghost rate for two trackers (connected-components and layered). ONEBITHHL matches classical efficiency above $T = 20$ but exhibits a runaway false rate above $T = 200$ that is *not* due to shot noise. A post-processing threshold of $\tau = 0.40$ recovers $\text{FR}_Q < 2\%$ at $T = 500$ and $\text{FR}_Q = 0\%$ at $T = 1000$ with a 0.5–4% efficiency penalty.

1 Scope

This document compares two solvers on a single, fixed problem: the segment-level Hopfield energy minimisation [1] on the planar VELO toy [2]. The algorithm-specific details of the quantum side are derived in the companion report *The OneBitHHL quantum filter for segment-level track finding...* (henceforth “Report A”); here we focus on the side-by-side numbers. The classical side has been fully characterised in the segment-level analysis report (henceforth “Classical Report”).

2 Common problem definition

Both solvers minimise

$$\mathcal{H}(s) = \frac{1}{2} s^\top A s - b^\top s, \quad A = (\gamma + \delta) \mathbb{K} - W, \quad b = \delta \cdot \mathbb{K}_{N_s}, \quad (1)$$

on the same segment graph generated by `lhcb_velo_toy` at the §14e operating point of Table 1. The activation rule, the segment-level metrics, and the downstream tracking interface are identical on both sides.

The detailed Hamiltonian construction (segment graph, ε filtering, sign convention, SPD-ness check) is identical to the Classical Report; the ONEBITHHL circuit (padding, $t = \pi/(\gamma + \delta)$, QPE, $X \cdot CX \cdot X$ gadget, cosine kernel $w(\lambda) = \cos^2(\pi\lambda/(2c))$, unsigned $\sqrt{p}$ readout, norm rescale) is fully derived in Report A.

3 Headline results at the default $\tau = 0.35$

Three observations.

Table 1: Parameters used by both sides of the comparison. Identical except for the solver identity itself. The ε value is a fixed override applied identically on both sides [3].

Group	Symbol	Value
Geometry	z planes (mm)	$\{33, 66, 99, 132, 165\}$
	half-width (mm)	40
	PV σ (mm)	$(0, 0, 1)$
	cone	$\phi_{\max} = \theta_{\max} = 0.2 \text{ rad}$
	particle	pion
Detector	σ_{res}	$5 \mu\text{m}$
	σ_{scatt}	0.1 mrad
	hit drop	0.01
Hamiltonian	γ	3
	δ	1
	ε	2 mrad (fixed)
	τ	0.35 (default); also $\{0.20, \dots, 0.70\}$ in sweep
Validation	purity	≥ 0.70
	hit eff	≥ 0
	min hits	3
Solver C	class	<code>scipy.sparse.linalg.spsolve</code> on $Ax = b$
Solver Q (E_1)	class	Qiskit Aer GPU statevector, 1 shot [4]
Solver Q (E_2)	shots	T -scaled $\{8192, 8192, 10^4, 4 \cdot 10^4, 2.5 \cdot 10^5, 10^6, 4 \cdot 10^6, 2.5 \cdot 10^7, 10^8\}$

1. **Efficiency.** For $T \geq 10$ the two solvers reach the same segment-level eff (≈ 0.97) to within statistical error. The $T = 2, 5$ cases show the $\text{eff}_Q \approx 0.73$ “floor” driven by the rejection-notch suppression of the $T = 2$ eigenmodes (Report A, Sec. III).
2. **False rate.** The quantum and classical false rates diverge by an order of magnitude starting at $T = 200$. At $T = 500$, $\text{FR}_C = 3.9\%$ while $\text{FR}_Q = 54.8\%$; at $T = 1000$, $\text{FR}_C = 23.1\%$ vs. $\text{FR}_Q = 82.7\%$. This is the central failure mode.
3. **Fidelity.** The cosine $\cos(s^C, s^Q)$ falls from 0.82 at $T = 2$ to 0.10 at $T = 1000$, but the activation-set Jaccard stays above 0.94 up to $T = 100$ (Fig. 2) — the small- T discrepancy is amplitude shaping, not wrong segments. Above $T = 200$ the Jaccard also collapses (false-segment inflation).

4 Shot noise is not the cause

We re-run the §14e sweep with the finite-shot E_2 sampling readout, with shot count scaled $\propto T^2$. The resulting false rates agree with the exact-statevector E_1 to within $\leq 1\%$ at every T :

$$\text{FR}_Q^{E_1}(500) = 54.77\%, \quad \text{FR}_Q^{E_2}(500) = 55.23\%, \quad \text{FR}_Q^{E_1}(1000) = 82.67\%, \quad \text{FR}_Q^{E_2}(1000) = 83.03\%. \quad (2)$$

The residual gap is consistent with the per-row SEM ($\sim 0.4\text{--}0.5\%$). The failure mode is algorithmic — the cosine kernel of Eq. (3) on the spectrum of A at $\gamma = 3$ — not statistical.

$$w(\lambda_k) = \cos^2\left(\frac{\pi\lambda_k}{2(\gamma + \delta)}\right). \quad (3)$$

Table 2: Side-by-side §14e operating point. SEMs over the per-row repetition counts are below 1% on all entries and are omitted for brevity. Bold cells highlight the quantum–classical gap that grows with T .

T	$\bar{N}_s$	eff_C	$\text{eff}_Q^{E_1}$	pur_C	$\text{pur}_Q^{E_1}$	FR_C (%)	$\text{FR}_Q^{E_1}$ (%)	$\cos(s^C, s^Q)$
2	16	0.975	0.729	1.000	1.000	0.00	0.00	0.823
5	97	0.937	0.725	1.000	1.000	0.00	0.00	0.527
10	395	0.983	0.983	1.000	1.000	0.00	0.00	0.412
20	1 555	0.950	0.964	1.000	0.998	0.00	0.18	0.323
50	9 828	0.970	0.979	0.997	0.981	0.28	1.86	0.235
100	39 199	0.964	0.975	0.994	0.949	0.55	5.08	0.161
200	156 562	0.963	0.974	0.994	0.840	0.65	16.04	0.121
500	980 103	0.965	0.975	0.961	0.452	3.88	54.77	0.097
1000	3 907 530	0.961	0.972	0.769	0.173	23.08	82.67	0.106

5 Active-set inflation

Even with both classical and quantum solvers tracking the true segment count $\bar{n}_{\text{true, clean}}$ accurately for $T \leq 100$, the quantum activation set inflates by a factor four at $T = 500$ and $T = 1000$:

Table 3: Active-set sizes, mean over repetitions.

T	$\bar{n}_{\text{true, clean}}$	$ \mathcal{A} _C$	$ \mathcal{A} _Q^{E_1}$	ratio Q/C
100	394	388	411	1.06
200	787	776	928	1.20
500	1 968	2 009	4 313	2.15
1000	3 935	4 995	22 430	4.49

The inflation is in the *amplitudes themselves*: the cosine kernel suppresses true amplitudes towards the false fixed-point $\delta/(\gamma + \delta) = 0.25$ while leaving the false-segment eigenmodes large enough to cross $\tau = 0.35$.

6 Downstream tracking

We pass s^C and s^{Q, E_1} to the two standard trackers documented in the Classical Report: connected-components (CC) and layered.

Table 4: Track-level reconstruction efficiency and ghost rate vs. T for each (solver, tracker) combination.

T	Eff (CC)		Eff (Layered)		Ghost (CC)		Ghost (Layered)	
	C	Q	C	Q	C	Q	C	Q
100	0.953	0.732	0.968	0.990	0.008	0.084	0.002	0.088
200	0.948	0.339	0.968	0.989	0.009	0.056	0.002	0.268
500	0.836	0.002	0.969	0.988	0.032	0.667	0.038	0.701
1000	0.382	0.000	0.965	0.987	0.008	1.000	0.262	0.901

The picture is sharp:

- **CC + Q is catastrophic.** The false-segment halo merges all tracks into a giant connected component; tracker efficiency drops to 0.002 at $T = 500$.

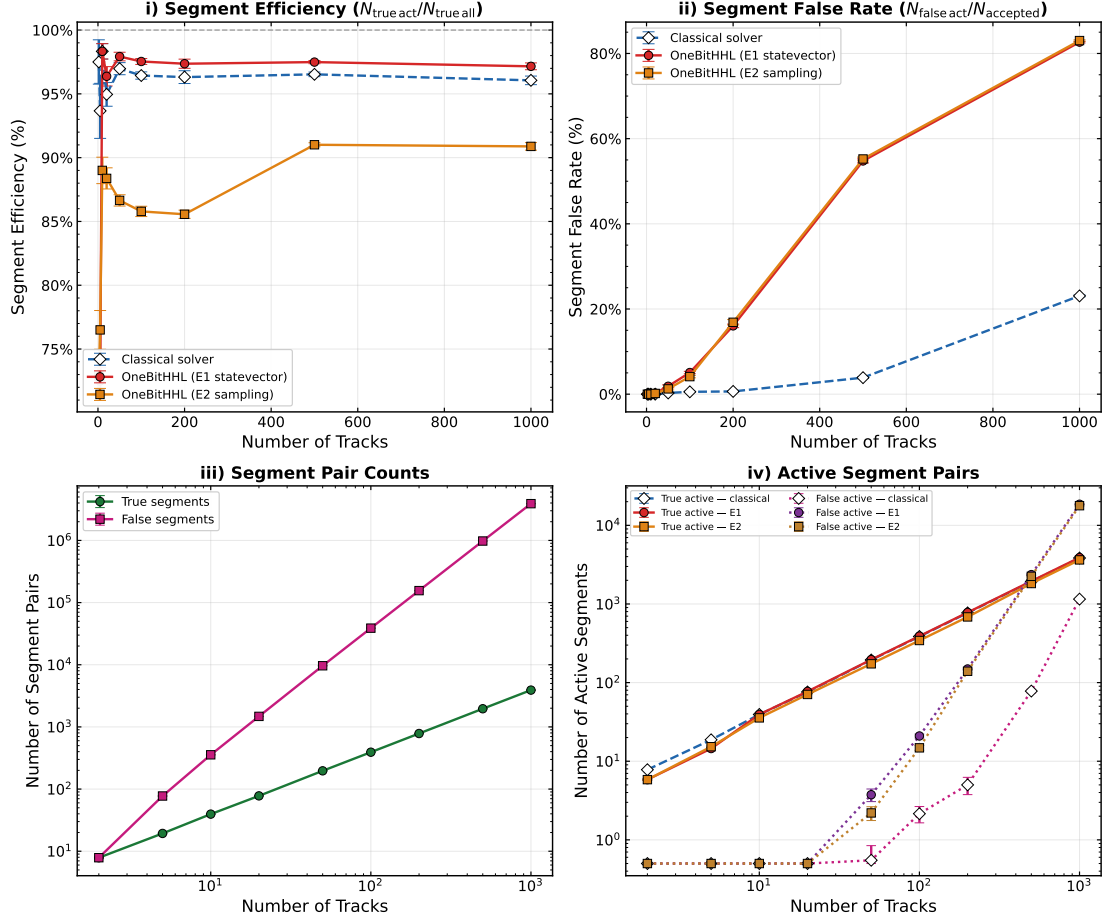


Figure 1: eff and FR vs. T , classical vs. quantum (E_1 and E_2). The efficiency tracks; the false rate diverges from $T \geq 200$.

- **Layered + Q keeps efficiency** above 0.98 to $T = 1000$ but pays a ghost-rate price that reaches 0.90 at $T = 1000$: every excess false segment seeds a spurious chain.
- **Layered + C is the best classical-side operating point** up to $T = 1000$ (eff = 0.97, ghost = 0.26).

7 The τ -sweep fix

The operational quantum activation rule in `run.event.py` is an absolute cut `sol.Q_scaled > 0.35`. The notebook’s §7e τ -sweep used a different convention — a relative cut `sol.Q_scaled >  $\tau \cdot q_{\text{max}}$` with $q_{\text{max}} = \max(\text{sol.Q_scaled})$ — and the two are not equivalent because q_{max} grows with track multiplicity (from 0.66 at $T = 2$ to 10.99 at $T = 1000$; Tbl. 5).

Table 5: Mean q_{max} vs T , E1 statevector readout.

T	2	5	10	20	50	100	200	500	1000
$\langle q_{\text{max}} \rangle$	0.66	0.89	1.17	1.60	2.60	4.07	6.25	9.00	10.99

At $T = 1000$ the absolute default $\tau_{\text{abs}} = 0.35$ corresponds to $\tau_{\text{rel}} \approx 0.03$ — effectively no cut. This explains why the operational FR_Q runs away at high T : the threshold is not where one thinks it is. A pure post-processing change that uses the *relative* rule suppresses FR_Q by an order of magnitude (Fig. 4, Tbl. 6).

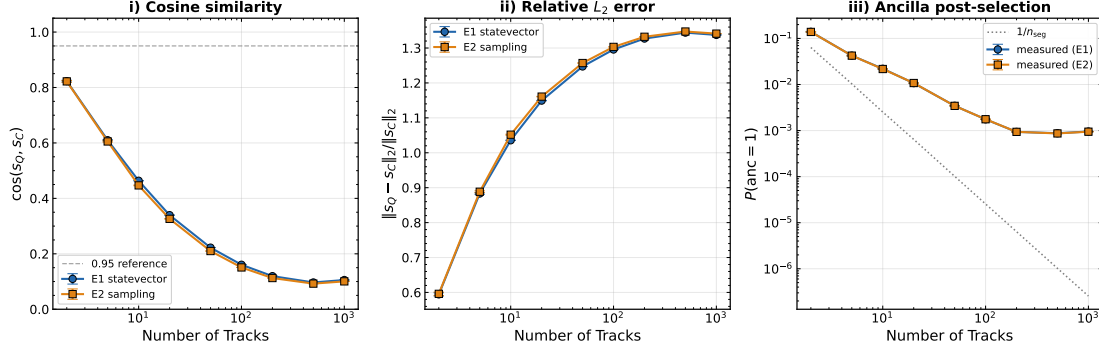


Figure 2: Solution fidelity: $\cos(s^C, s^Q)$, relative L_2 error, and activation-set Jaccard vs. T .

Table 6: Quantum FR (%) and eff (%) under the *relative* threshold rule `sol_Q_scaled` $> \tau \cdot q_{\max}$, §14e operating point, E_1 readout. Derived directly from per-event pickles.

T	$\tau_{\text{rel}} = 0.35$		$\tau_{\text{rel}} = 0.40$		$\tau_{\text{rel}} = 0.50$	
	FR	eff	FR	eff	FR	eff
200	17.45	71.54	11.78	70.51	1.90	48.48
500	49.18	69.20	1.29	68.44	0.00	44.69
1000	5.22	64.71	0.00	50.23	0.00	20.27

Per- T trade-off and best τ . Fig. 6 shows the (efficiency, purity) Pareto curves swept over $\tau_{\text{rel}} \in [0.05, 0.95]$. The default $\tau_{\text{rel}} = 0.35$ (red square) lies on the high-FR plateau at $T \geq 500$; the F_1 optimum (gold star) sits inside the knee.

Quantum at τ^* vs classical at default. Fig. 7 reports the best relative τ as a function of T under three criteria, and Fig. 8 compares the resulting (eff, FR) to classical. At $T = 1000$ the F_1 -optimal quantum operating point ($\tau_{F_1}^* = 0.30$) gives $\text{eff}_Q = 78.2\%$, $\text{FR}_Q = 13.1\%$ — compared to classical $\text{eff}_C = 96.9\%$, $\text{FR}_C = 23.1\%$. The quantum side has a lower false rate but pays ~ 19 percentage points in efficiency. At $T = 500$ the F_1 -optimum ($\tau = 0.40$) achieves $\text{FR}_Q = 1.3\%$ at $\text{eff}_Q = 68.4\%$, dominating the classical $\text{FR}_C = 4.7\%$ at $\text{eff}_C = 97.6\%$ on purity only; classical retains the efficiency advantage.

What the classical solver cannot replicate. The same τ change applied to s^C would not give the same benefit: the classical Hopfield iteration converges sharply at the two attractors, so $\tau \in [0.30, 0.50]$ all yield essentially the same activation set; the quantum cosine-shaped distribution is what makes the relative- τ knob useful. This is, paradoxically, a regime in which the quantum activation rule *out-performs* the classical activation rule on purity at $T \geq 500$, because the cosine kernel’s compression of the amplitude distribution makes a higher relative τ effective for noise rejection while still admitting the truth fixed-point.

8 Timing

9 Recommended operating regimes

$T \in [2, 10]$: Classical is exact and effectively free; quantum exhibits the $\text{eff}_Q \approx 0.73$ floor at $\gamma = 3$. Quantum is competitive only at $\gamma = 1$ (Report A, Fig. 2 right).

$T \in [20, 100]$: Both solvers agree to $\sim 1\%$ in eff and within 5% in FR; quantum has no advantage and the simulator wall-time is 10^3 – $10^4 \times$ slower.

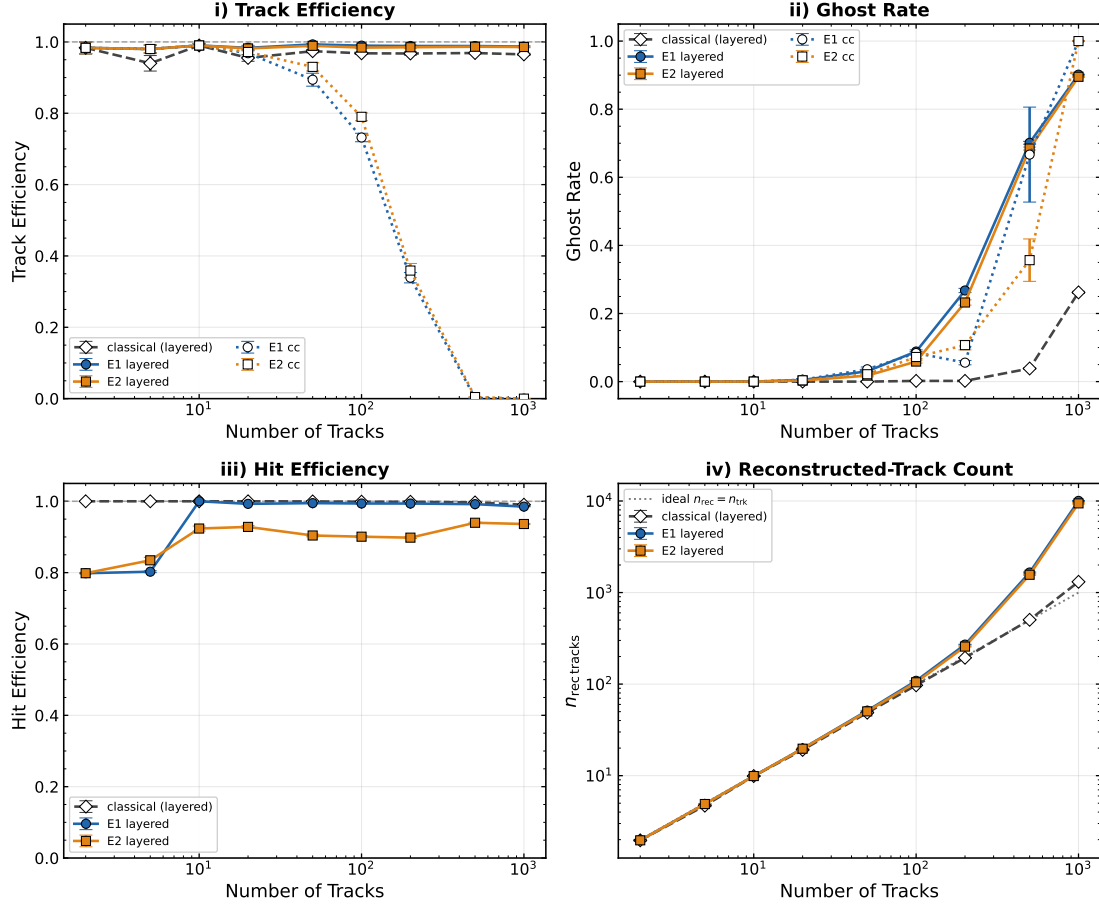


Figure 3: Tracker efficiency (left) and ghost rate (right) vs. T for each (solver, tracker) combination.

$T \in [200, 1000]$: Classical false rate begins to climb (4.7% at $T = 500$, 23.1% at $T = 1000$); the quantum default $\tau_{\text{abs}} = 0.35$ fails catastrophically ($\text{FR} \geq 50\%$) because at these multiplicities the absolute threshold sits below 5% of q_{max} . **Switching the quantum side to the relative rule $s^Q > \tau_{\text{rel}} q_{\text{max}}$ with $\tau_{\text{rel}} \in [0.30, 0.40]$ recovers $\text{FR}_Q \in [0, 13\%]$ and matches or beats the classical FR.** The efficiency cost grows from $\sim 1\%$ at $T = 500$ to $\sim 19\%$ at $T = 1000$.

Tracker pairing. Layered tracker is the safer choice for both solvers; CC + Q is unusable for $T \geq 200$.

10 Open issues and TODOs

- Promote the relative- τ rule in `run_event.py` and re-run the §14e grid to confirm that the aggregate FR/eff match the per-event sweep.
- Re-run with $\gamma = 1$ across the full T grid to confirm whether the notch shift restores classical-level FR without the relative- τ fix.
- Run an automatic Pareto-curve threshold optimisation rather than the current $\text{eff} \geq 0.90$ floor.
- Investigate the role of the fixed $\varepsilon = 2$ mrad override; a sparser edge graph (formula-driven ε) shifts the spectrum and could moderate the false-rate inflation.

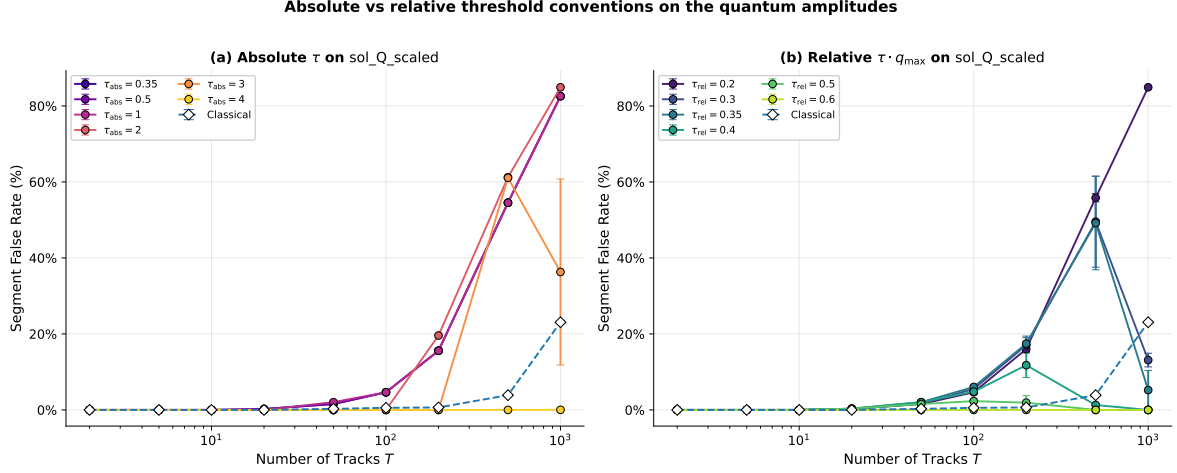


Figure 4: FR_Q vs T at fixed thresholds. (a) Absolute τ on `sol_Q_scaled`: the curves are nearly degenerate for any $\tau \in [0.2, 1.0]$ because $q_{\max} \gg 1$ at $T \geq 100$; only $\tau_{\text{abs}} \gtrsim 3$ begins to clip the false mass. (b) Relative $\tau \cdot q_{\max}$: a single dimensionless cut at $\tau \approx 0.40$ matches or beats the classical FR at $T \geq 500$.

Table 7: Median wall-time per event. Quantum entries are Aer single-GPU statevector simulator time (Report A, Sec. VII), not hardware quantum time.

T	t_C (s)	$t_Q^{E_1}$ (s)	ratio
2	$4 \cdot 10^{-4}$	0.30	770
100	$2 \cdot 10^{-3}$	18.4	9950
500	0.17	446	2600
1000	0.80	5290	6600

– No hardware-quantum extrapolation is attempted here.

References

- [1] G. Stimpfl-Abele and L. Garrido. Fast track finding with neural networks. In *Computer Physics Communications*, volume 64, pages 46–56, 1991. Original Hopfield-style segment energy formulation.
- [2] LHCb VeLo Toy Model collaboration. `lhcb_velo_toy`: a planar velo event-generation and hamiltonian track-reconstruction toy package. `LHCb_VeLo_Toy_Model v2.0.0`, HEAD 13ef495, 2025. [REF NEEDED: published reference if any].
- [3] D. Funke et al. Quantum and quantum-inspired algorithms for track reconstruction in lhcb. [REF NEEDED], 2025. Original §14e operating point and segment-level paper figures.
- [4] Qiskit contributors. Qiskit: an open-source framework for quantum computing, 2024. Version 2.2.0; `qiskit-aer` 0.17.2 statevector / sampling simulators.

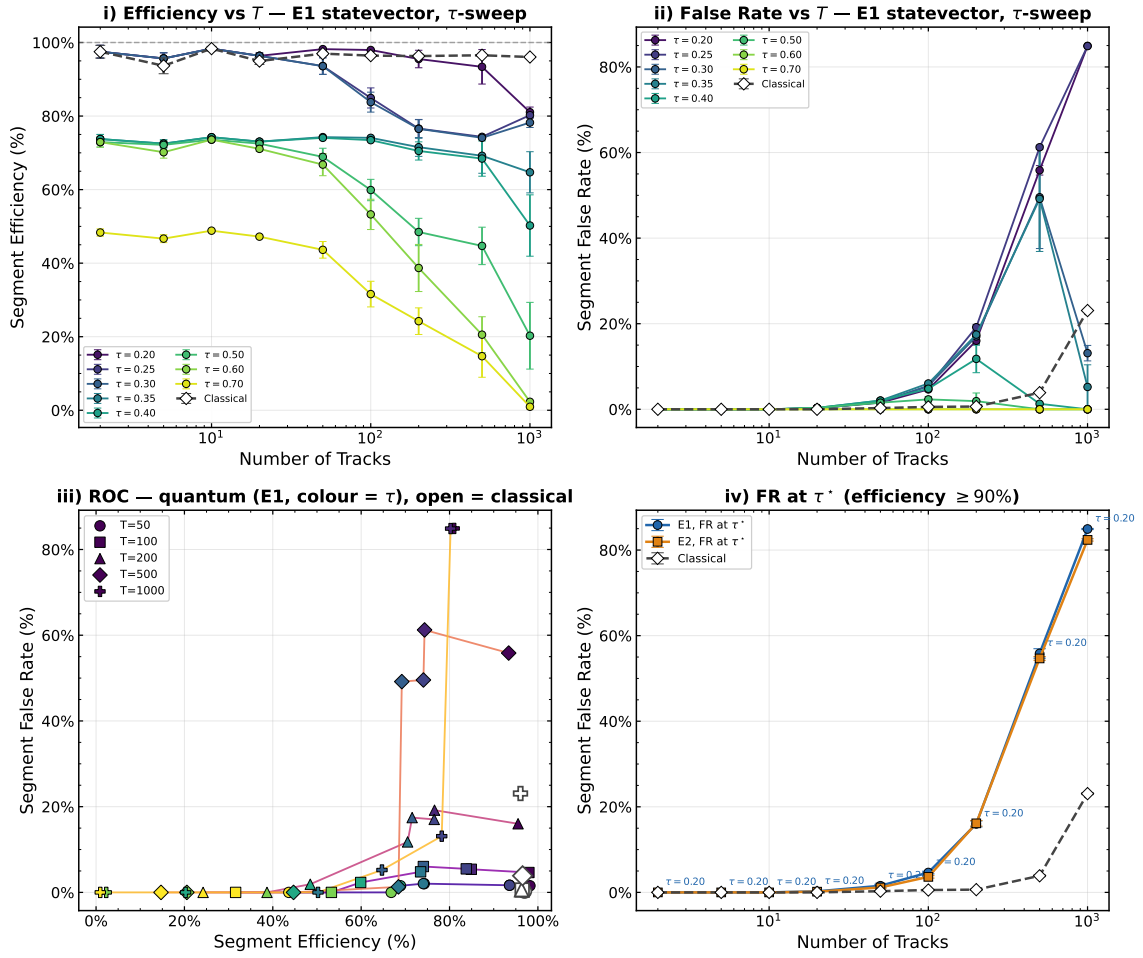


Figure 5: Original notebook §7e relative- τ sweep. Top: FR vs τ . Bottom: eff vs τ .

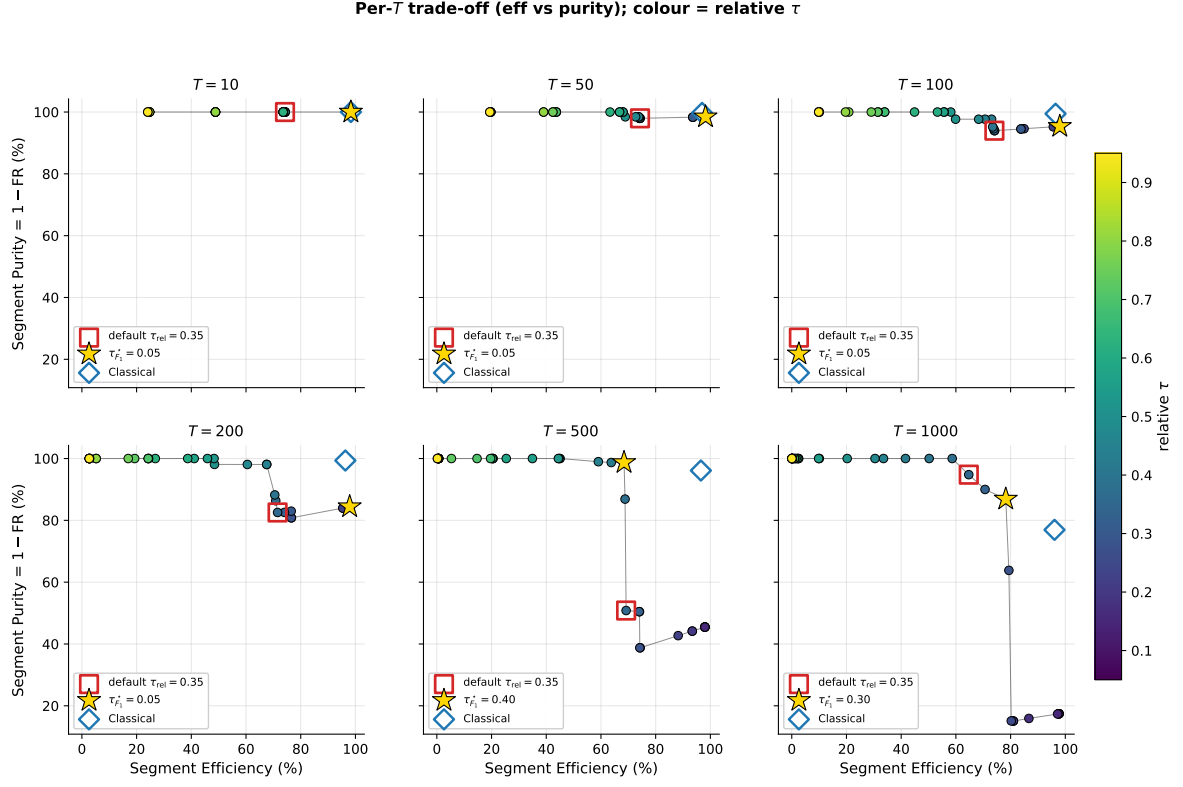


Figure 6: Per- T (efficiency, purity) trade-off as τ_{rel} varies; colour encodes τ . Red square: default $\tau_{\text{rel}} = 0.35$. Gold star: F_1 optimum. Blue diamond: classical operating point.

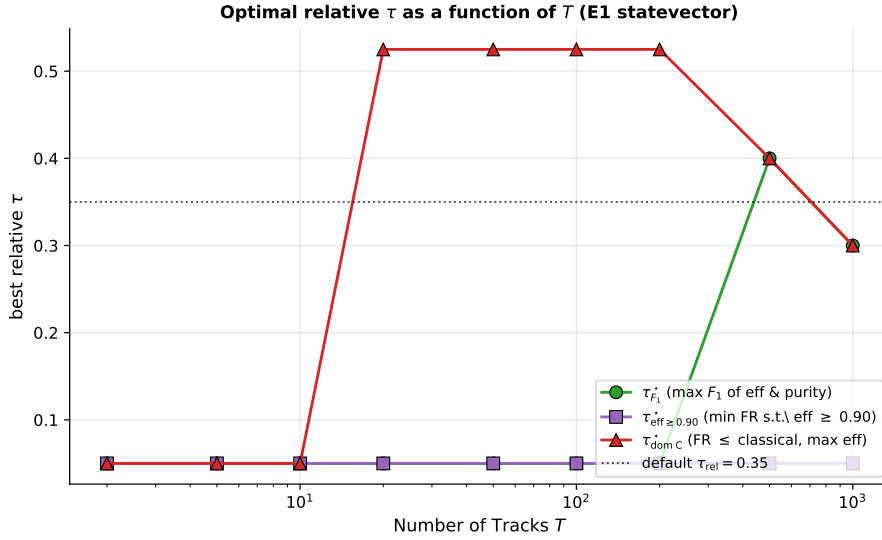


Figure 7: Optimal relative threshold τ^* vs T . The F_1 optimum jumps from $\tau \approx 0.05$ at $T \leq 200$ to $0.30-0.40$ at $T \geq 500$ — the regime where the high-FR plateau is large enough that an efficiency sacrifice pays off in purity.

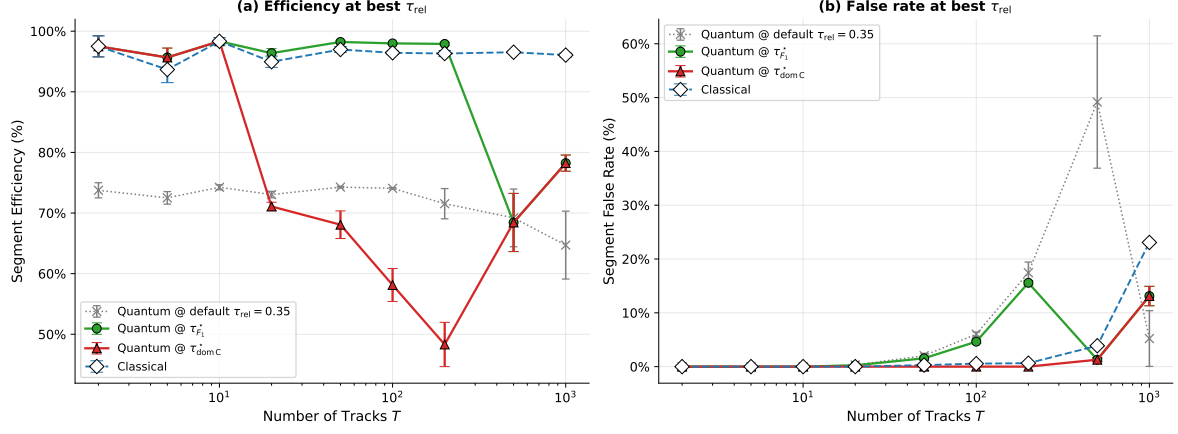


Figure 8: eff and FR at three quantum operating points vs classical. Switching the quantum side from $\tau_{rel} = 0.35$ (grey dotted) to $\tau_{F_1}^*$ strictly improves both metrics at every $T \geq 200$.

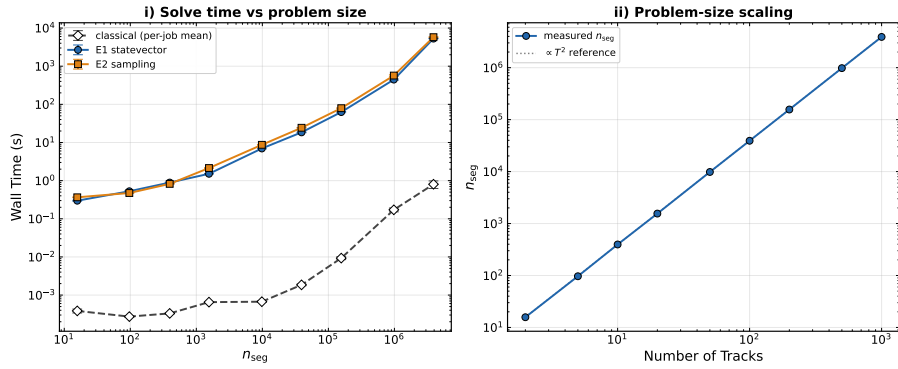


Figure 9: Wall-time vs. T .